# Quantum Computing, Quantum Machine Learning and Quantum Information Theories

Morten Hjorth-Jensen[1]

Department of Physics, University of Oslo, Norway[1]

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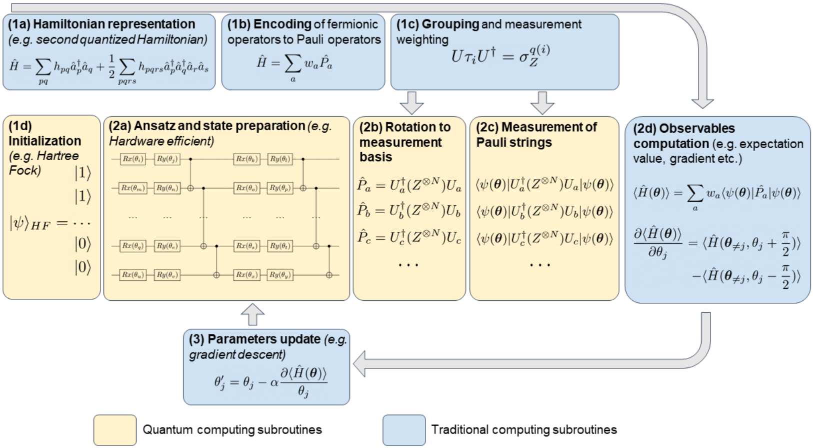
# Plans for the week of February 23-27

- Repetition from last week on the VQE with applications to a one-qubit problem
- Setting up calculations with the Variational Quantum Eigensolver (VQE) for one- and two-qubits Hamiltonians
- Deriving the expressions for the gradients and setting up measurements in the computational basis
- Video of lecture at `https://youtu.be/MVLbBcTPwqg`
- Whiteboard notes at `https://github.com/CompPhysics/QuantumComputingMachineLearning/blob/gh-pages/doc/HandWrittenNotes/2026/Notesweek6.pdf`

# Reading suggestions

- Hundt's text section 6.11 on the VQE
- Jupyter-notebook on VQE and single-qubit problem at https://github.com/CompPhysics/QuantumComputingMachineLearning/blob/gh-pages/doc/pub/week6/ipynb/single_qubit_vqe_modified.ipynb
- Jupyter-notebook on VQE and two-qubit problem at https://github.com/CompPhysics/QuantumComputingMachineLearning/blob/gh-pages/doc/pub/week6/ipynb/two_qubit_vqe_modified.ipynb

# VQE overview



# One-qubit Hamiltonian

Last week we simply computed the expectations values of the various Pauli matrices by evaluating expressions like

$$\langle\psi(\theta)|\boldsymbol{X}|\psi(\theta)\rangle,$$

for the Pauli-$\boldsymbol{X}$ matrix and similarly for the other Pauli matrices and the $2 \times 2$ identity matrix $\boldsymbol{I}$. This is however not the way we end up evaluating such expectation values on an actual quantum computer. We have to transform the set of measurements to be performed to the computational basis we have chosen.

# Our computational basis

The computational basis we have chosen is given by the eigenvectors of the Pauli-$\boldsymbol{Z}$ matrix, namely $|0\rangle$ and $|1\rangle$.
For the actual implementation of these measurements we need three different circuits where

- $\boldsymbol{Z}$-measurement: Direct measurement.
- $\boldsymbol{X}$-measurement: Apply a Hadamard gate $\boldsymbol{H}$, then measure $\boldsymbol{Z}$.
- $\boldsymbol{Y}$-measurement: Apply $\boldsymbol{S}^\dagger\boldsymbol{H}$ gates, then measure $\boldsymbol{Z}$.

# Shot-Based Sampling

## Finite measurement shots introduce statistical noise

- Each measurement yields 0 or 1 with probabilities $|\alpha|^2$ and $|\beta|^2$
- Expectation values estimated from shot counts
- Error scales as $\mathcal{O}(1/\sqrt{N_{\text{shots}}})$

The codes we discussed last week did not include a shot-based sampling.

# Parameter-shift rule for evaluating gradients

In our discussions last week we used a quasi-Newton methods to find the roots of the gradients, meaning that we evaluated the gradients by finite difference methods.
We will now derive a more efficient way of evaluating the derivatives of the energy with respect to the parameters of our circuit (the rotation angles).

# Gradient descent and calculations of gradients

In order to optimize the VQE ansatz, we need to compute derivatives with respect to the variational parameters. Here we develop first a simpler approach tailored to the one-qubit case. For this particular case, we have defined an ansatz in terms of the Pauli rotation matrices.

# Setting up gradients

These define an arbitrary one-qubit state on the Bloch sphere through the expression

$$|\psi\rangle = |\psi(\theta,\phi)\rangle = R_y(\phi)R_x(\theta)|0\rangle.$$

Each of these rotation matrices can be written in a more general form as

$$R_i(\gamma) = \exp -(\imath\frac{\gamma}{2}\sigma_i) = \cos(\frac{\gamma}{2})\boldsymbol{I} - \imath\sin(\frac{\gamma}{2})\boldsymbol{\sigma}_i,$$

where $\sigma_i$ is one of the Pauli matrices $\sigma_{x,y,z}$.

# Derivatives

It is easy to see that the derivative with respect to $\gamma$ is

$$\frac{\partial R_i(\gamma)}{\partial \gamma} = -\frac{\imath}{2}\boldsymbol{\sigma}_i R_i(\gamma).$$

# Derivatives of the expectation value of the Hamiltonian

We can now calculate the derivative of the expectation value of the Hamiltonian in terms of the angles $\theta$ and $\phi$. We have two derivatives

$$\frac{\partial}{\partial\theta}\left[\langle\psi(\theta,\phi)|\boldsymbol{H}|\psi(\theta,\phi)\rangle\right] = \frac{\partial}{\partial\theta}\left[\langle\boldsymbol{H}(\theta,\phi)\rangle\right] = \langle\psi(\theta,\phi)|\boldsymbol{H}(-\frac{\imath}{2}\boldsymbol{\sigma}_x|\psi(\theta,\phi)\rangle+$$

and

$$\frac{\partial}{\partial\phi}\left[\langle\psi(\theta,\phi)|\boldsymbol{H}|\psi(\theta,\phi)\rangle\right] = \frac{\partial}{\partial\phi}\left[\langle\boldsymbol{H}(\theta,\phi)\rangle\right] = \langle\psi(\theta,\phi)|\boldsymbol{H}(-\frac{\imath}{2}\boldsymbol{\sigma}_y|\psi(\theta,\phi)\rangle-$$

# Two addtional expectation values

This means that we have to calculate two additional expectation values in addition to the expectation value of the Hamiltonian itself. If we stay with an ansatz for the single qubit states given by the above rotation operators, we can, following for example the article by Maria Schuld et al, show that the derivative of the expectation value of the Hamiltonian can be written as (we focus only on a given angle $\phi$)

$$\frac{\partial}{\partial\phi}\left[\langle \boldsymbol{H}(\phi)\rangle\right] = \frac{1}{2}\left[\langle \boldsymbol{H}(\phi+\frac{\pi}{2})\rangle - \langle \boldsymbol{H}(\phi-\frac{\pi}{2})\rangle\right].$$

# Rotations again and again

To see this, consider again the definition of the rotation operators. We can write these operators as

$$R_i(\phi) = \exp -\frac{\imath}{2}(\phi \boldsymbol{\sigma}_i),$$

with $\boldsymbol{\sigma}_i$, with $\boldsymbol{\sigma}_i$ being any of the Pauli matrices $\boldsymbol{X}$, $\boldsymbol{Y}$ and $\boldsymbol{Z}$. The latter can be generalized to other unitary matrices as well. The derivative with respect to $\phi$ gives

$$\frac{\partial R_i(\phi)}{\partial \phi} = -\frac{\imath}{2}\boldsymbol{\sigma}_i \exp -\imath(\phi \boldsymbol{\sigma}_i) = -\frac{\imath}{2}\boldsymbol{\sigma}_i R_i(\phi).$$

# Bloch sphere math

Our ansatz for a general one-qubit state on the Bloch sphere contains the product of a rotation around the $x$-axis and the $y$-axis. In the derivation here we focus only on one angle however. Our ansatz is then given by

$$|\psi\rangle = R_i(\phi)|0\rangle,$$

and the expectation value of our Hamiltonian is

$$\langle\psi|\mathcal{H}|\psi\rangle = \langle 0|R_i(\phi)^\dagger \mathcal{H} R_i(\phi)|0\rangle.$$

# Derivatives

Our derivative with respect to the angle $\phi$ has a similar structure, that is

$$\frac{\partial}{\partial\phi}\left[\langle\psi(\theta,\phi)|\boldsymbol{H}|\psi(\theta,\phi)\rangle\right] = \langle\psi(\theta,\phi)|\boldsymbol{H}(-\frac{\imath}{2}\boldsymbol{\sigma}_y|\psi(\theta,\phi)\rangle + \text{ h.c.}$$

# Rewriting

In order to rewrite the equation of the derivative, the following relation is useful

$$\langle\psi|\hat{A}^\dagger\hat{B}\hat{C}|\psi\rangle = \frac{1}{2}\Big[\langle\psi|(\hat{A}+\hat{C})^\dagger\hat{B}(\boldsymbol{A}+\hat{C})|\psi\rangle - \langle\psi|(\hat{A}-\hat{C})^\dagger\hat{B}(\boldsymbol{A}-\hat{C})|\psi\rangle$$

where $\hat{A}$, $\hat{B}$ and $\hat{C}$ are arbitrary hermitian operators.

# Final manipulations

If we identify these operators as $\hat{A} = \boldsymbol{I}$, with $\boldsymbol{I}$ being the unit operator, $\hat{B} = \mathcal{H}$ our Hamiltonian, and $\hat{C} = -\imath\boldsymbol{\sigma}_i/2$, we obtain the following expression for the expectation value of the derivative (excluding the hermitian conjugate)

$$\langle\psi|\boldsymbol{I}^{\dagger}\mathcal{H}(-\frac{\imath}{2}\boldsymbol{\sigma}_i|\psi\rangle = \frac{1}{2}\left[\langle\psi|(\boldsymbol{I}-\frac{\imath}{2}\boldsymbol{\sigma}_i)^{\dagger}\mathcal{H}(\boldsymbol{I}-\frac{\imath}{2}\boldsymbol{\sigma}_i)|\psi\rangle - \langle\psi|(\boldsymbol{I}+\frac{\imath}{2}\boldsymbol{\sigma}_i)^{\dagger}\mathcal{H}(\right.$$

# The expressions to implement

If we then use that the rotation matrices can be rewritten as

$$R_i(\phi) = \exp -(\imath\frac{\phi}{2}\sigma_i) = \cos(\frac{\phi}{2})\boldsymbol{I} - \imath\sin(\frac{\phi}{2})\boldsymbol{\sigma}_i,$$

we see that if we set the angle to $\phi = \pi/2$, we have

$$R_i(\frac{\pi}{2}) = \cos(\frac{\pi}{4})\boldsymbol{I} - \imath\sin(\frac{\pi}{4})\boldsymbol{\sigma}_i = \frac{1}{\sqrt{2}}\left(\boldsymbol{I} - \frac{\imath}{2}\boldsymbol{\sigma}_i\right).$$

# Final expression

This means that we can write

$$\langle\psi|\boldsymbol{I}^{\dagger}\mathcal{H}(-\frac{\imath}{2}\boldsymbol{\sigma}_i|\psi\rangle = \frac{1}{2}\left[\langle\psi|R_i(\frac{\pi}{2})^{\dagger}\mathcal{H}R_i(\frac{\pi}{2})|\psi\rangle - \langle\psi|R_i(-\frac{\pi}{2})^{\dagger}\mathcal{H}R_i(-\frac{\pi}{2})^{\dagger}|\psi\right.$$

# Basics of gradient descent and stochastic gradient descent

In order to implement the above equations, we need to remind the reader about basic elements of various optimization approaches. Our main focus here will be various gradient descent approaches and quasi-Newton methods like Broyden's algorithm and variations thereof.

This material is covered by the lectures from FYS4411 on gradient optimization

# Two-qubit Hamiltonian

Here we discuss how to rewrite the two-qubit Hamiltonian defined by the following Hamiltonian matrix (project 1)

$$\mathcal{H} = \begin{bmatrix} \epsilon_1 + V_z & 0 & 0 & V_x \\ 0 & \epsilon_2 - V_z & V_x & 0 \\ 0 & H_x & \epsilon_3 - V_z & 0 \\ H_x & 0 & 0 & \epsilon_4 + V_z \end{bmatrix}.$$

This Hamiltonian can be rewritten in terms of various one-qubit matrices.

# Definitions

We define

$$\epsilon_{II} = \frac{\epsilon_1 + \epsilon_2 + \epsilon_3 + \epsilon_4}{4},$$

$$\epsilon_{ZI} = \frac{\epsilon_1 + \epsilon_2 - \epsilon_3 - \epsilon_4}{4},$$

$$\epsilon_{IZ} = \frac{\epsilon_1 - \epsilon_2 + \epsilon_3 - \epsilon_4}{4},$$

$$\epsilon_{ZZ} = \frac{\epsilon_1 - \epsilon_2 - \epsilon_3 + \epsilon_4}{4}.$$

# The Hamiltonian in terms of Pauli-$\boldsymbol{X}$ and Pauli-$\boldsymbol{Z}$ matrices

With these definitions we can rewrite our two-qubit Hamiltonian as

$$\mathcal{H} = \mathcal{H}_0 + \mathcal{H}_I$$

with

$$\mathcal{H}_0 = \epsilon_{II}\boldsymbol{I}\otimes\boldsymbol{I} + \epsilon_{ZI}\boldsymbol{Z}\otimes\boldsymbol{I} + \epsilon_{IZ}\boldsymbol{I}\otimes\boldsymbol{Z} + \epsilon_{ZZ}\boldsymbol{Z}\otimes\boldsymbol{Z},$$

and

$$\mathcal{H}_I = V_z\boldsymbol{Z}\otimes\boldsymbol{Z} + V_x\boldsymbol{X}\otimes\boldsymbol{X}.$$

# How do we perform measurements?

The above tensor products have to be rewritten in terms of specific transformations so that we can perform the measurements in the basis of the Pauli-$\boldsymbol{Z}$ matrices. As we discussed earlier, we need to find a transformation of the form

$$\mathcal{P} = \boldsymbol{U}^{\dagger}\boldsymbol{M}\boldsymbol{U},$$

where $\mathcal{P}$ represents some combination of the Pauli matrices and the identity matrix, $\boldsymbol{U}$ is a unitary matrix and $\boldsymbol{M}$ represents the gate/matrix which performs the measurements, often represented by a Pauli-$\boldsymbol{Z}$ gate/matrix.

# Rewriting our strings of Pauli matrices

**Note:** The derivations here follow Hundt's text section 6.11. See additional notes and the jupyter-notebooks on how to perform measurements on both qubits.
As discussed last week and reviewed above, to perform a measurement, it is often useful to rewrite the string of Pauli matrices in a specific order or to simplify the expression.
Consider a string of Pauli matrices of the form:

$$P = \sigma_{i_1} \otimes \sigma_{i_2} \otimes \cdots \otimes \sigma_{i_n},$$

where each $\sigma_{i_k}$ is one of the Pauli matrices $\boldsymbol{X}$, $\boldsymbol{Y}$, $\boldsymbol{Z}$, or the identity matrix $\boldsymbol{I}$.

# Rewriting the string of matrices

To rewrite this string for measurement purposes, follow these steps:

## Commute and reorder:

Use the commutation relations of the Pauli matrices to reorder the string. The commutation relations are:

$$[\sigma_i, \sigma_j] = 2i\epsilon_{ijk}\sigma_k,$$

where $\epsilon_{ijk}$ is the Levi-Civita symbol. This allows you to move certain Pauli matrices to the left or right in the string.

## Simplify using identities:

Use the fact that $\sigma_i^2 = I$ and the above mentioned commutation relation.

# More rewriting tricks

## Group terms:

Group terms that are easier to measure together. For example, if you have a term like $\boldsymbol{X} \otimes \boldsymbol{X}$, you can measure both qubits in the $\boldsymbol{X}$-basis simultaneously.

## Diagonalize if necessary:

If the final expression is not diagonal, you may need to apply a unitary transformation to diagonalize it before measurement. For example,to measure $\boldsymbol{X}$, you can apply the Hadamard gate $\boldsymbol{H}$ to transform it into $\boldsymbol{X}$:

$$\boldsymbol{HXH} = \boldsymbol{Z}.$$

# The $\boldsymbol{Z} \otimes \boldsymbol{I}$ term

The explicit matrix is

$$\boldsymbol{Z} \otimes \boldsymbol{I} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}.$$

When we perform a measurement on the first qubit, we see that this matrix gives us the correct eigenvalues for the first qubit (but not for the second one). To see this multiply the above matrix with our computational basis states, that is the states $|00\rangle = |0\rangle \times |0\rangle$, $|01\rangle$, $|10\rangle$ and $|11\rangle$.
This process is referred to in the language of Pauli measurements as **measuring Pauli-Z** and is entirely equivalent to performing a computational basis measurement.

# The specific eigenvalues

Multiplying with the $|00\rangle$ state we get

$$\boldsymbol{Z} \otimes \boldsymbol{I} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix},$$

and

$$\boldsymbol{Z} \otimes \boldsymbol{I} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} = -1 \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}.$$

We don't get the correct eigenvalues if we perform the measurement on the second qubit!

# The $\boldsymbol{I} \otimes \boldsymbol{Z}$ term

Our strategy is to rewrite all the strings of Pauli operators via specific unitary transformations in order to obtain a final operator $\boldsymbol{P} = \boldsymbol{Z}_1 \otimes \boldsymbol{I}_2 \otimes \boldsymbol{I}_3 \otimes \cdots \otimes \boldsymbol{I}_N$, where the subscripts indicate the various qubits. We can then perform the measurement on the first qubit only.
We can rewrite $\boldsymbol{I} \otimes \boldsymbol{Z}$ via the SWAP gate

$$\mathrm{SWAP} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

# More on the $\boldsymbol{I} \otimes \boldsymbol{Z}$ term

We have then that

$$\boldsymbol{P} = \mathrm{SWAP}^{\dagger}(\boldsymbol{I} \otimes \boldsymbol{Z})\mathrm{SWAP} = \boldsymbol{Z} \otimes \boldsymbol{I}.$$

We note that the original $\boldsymbol{I} \otimes \boldsymbol{Z}$ does not give the correct eigenvalues when measured on the first qubit. Try this as an exercise.

## The $\boldsymbol{Z} \otimes \boldsymbol{Z}$ term

Thus the tensor products of two Pauli-$\boldsymbol{Z}$ operators forms a matrix composed of two spaces consisting of $+1$ and $-1$ as eigenvalues. As with the single-qubit case, both constitute a half-space, meaning that half of the accessible vector space belongs to the eigenspace with eigenvalue $+1$ and the remaining half to the eigenspace with eigenvalue $-1$.

This term gives the correct eigenvalue when operating on the first qubit. In principle thus we don't need to rewrite string of operators. However, as discussed below as well, let us rewrite it via a unitary transformation in order to have $\boldsymbol{P} = \boldsymbol{Z} \otimes \boldsymbol{I}$. To do so, consider the transformation

$$\boldsymbol{P} = CX_{10}^{\dagger}(\boldsymbol{Z} \otimes \boldsymbol{Z})CX_{10},$$

where we have

$$\mathrm{CX}_{10} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}.$$

# Performing the transformation

Performing the operation gives

$$\boldsymbol{P} = CX_{10}^{\dagger}(\boldsymbol{Z} \otimes \boldsymbol{Z})CX_{10} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix} = \boldsymbol{Z} \otimes \boldsymbol{I},$$

which allows us to transform the original tensor product $\boldsymbol{Z} \otimes \boldsymbol{Z}$ into a matrix where we perform the measurement in the basis of the first qubit.
To see this, act with $\boldsymbol{P}$ on the states $|00\rangle = |0\rangle \times |0\rangle$, $|01\rangle$, $|10\rangle$ and $|11\rangle$.

# Transformations

Any unitary transformation of such matrices also describes two half-spaces labeled with eigenvalues. For example

$$\boldsymbol{X} \otimes \boldsymbol{X} = \boldsymbol{H} \otimes \boldsymbol{H}(\boldsymbol{Z} \otimes \boldsymbol{Z})\boldsymbol{H} \otimes \boldsymbol{H}, ,$$

follows from from the identity that $\boldsymbol{Z} = \boldsymbol{HXH}$. Similar to the one-qubit case, all two-qubit Pauli-measurements can be written in terms of unitary transformations $\boldsymbol{U}^\dagger(\boldsymbol{Z} \otimes \boldsymbol{I})\boldsymbol{U}$ with $\boldsymbol{U}$ being $4 \times 4$ unitary matrices.

# More terms

Let us look at the $\boldsymbol{X} \otimes \boldsymbol{X}$ term in the simpler two-qubit Hamiltonian.
This term can be rewritten as

$$\boldsymbol{P} = (CX_{10}(\boldsymbol{H} \otimes \boldsymbol{H}))^{\dagger}(\boldsymbol{X} \otimes \boldsymbol{X})(CX_{10}(\boldsymbol{H} \otimes \boldsymbol{H})),$$

which results in

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}.$$

We recognize this result as our $\boldsymbol{Z} \otimes \boldsymbol{I}$ tensor product.

# Explicit expressions

In order to perform our measurements for our simple two-qubit Hamiltonian we need the following unitary transformations $\boldsymbol{U}$

$$
\begin{array}{rl}
\boldsymbol{Z} \otimes \boldsymbol{I} & \qquad \boldsymbol{U} = \boldsymbol{I} \otimes \boldsymbol{I} \\
\boldsymbol{I} \otimes \boldsymbol{Z} & \qquad \boldsymbol{U} = \mathrm{SWAP} \\
\boldsymbol{Z} \otimes \boldsymbol{Z} & \qquad \boldsymbol{U} = CX_{10} \\
\boldsymbol{X} \otimes \boldsymbol{X} & \qquad \boldsymbol{U} = CX_{10}(\boldsymbol{H} \otimes \boldsymbol{H})
\end{array}
$$

These are the gates that are relevant for the simpler two-qubit Hamiltonian of project 1.

# More complete list and derivations of expressions for strings of operators

For a two qubit system we list here the possible transformations

$$
\begin{array}{ll}
\boldsymbol{Y} \otimes \boldsymbol{I} & \boldsymbol{U} = \boldsymbol{HS}^{\dagger} \otimes \boldsymbol{I} \\
\boldsymbol{I} \otimes \boldsymbol{X} & \boldsymbol{U} = (\boldsymbol{H} \otimes \boldsymbol{I})\mathrm{SWAP} \\
\boldsymbol{I} \otimes \boldsymbol{X} & \boldsymbol{U} = (\boldsymbol{H} \otimes \boldsymbol{I})\mathrm{SWAP} \\
\boldsymbol{I} \otimes \boldsymbol{X} & \boldsymbol{U} = (\boldsymbol{HS}^{\dagger} \otimes \boldsymbol{I})\mathrm{SWAP} \\
\boldsymbol{X} \otimes \boldsymbol{Z} & \boldsymbol{U} = CX_{10}(\boldsymbol{H} \otimes \boldsymbol{I}) \\
\boldsymbol{Y} \otimes \boldsymbol{Z} & \boldsymbol{U} = CX_{10}(\boldsymbol{HS}^{\dagger} \otimes \boldsymbol{I}) \\
\boldsymbol{Z} \otimes \boldsymbol{Y} & \boldsymbol{U} = CX_{10}(\boldsymbol{I} \otimes \boldsymbol{HS}^{\dagger}) \\
\boldsymbol{Y} \otimes \boldsymbol{X} & \boldsymbol{U} = CX_{10}(\boldsymbol{HS}^{\dagger} \otimes \boldsymbol{H}) \\
\boldsymbol{X} \otimes \boldsymbol{Y} & \boldsymbol{U} = CX_{10}(\boldsymbol{H} \otimes \boldsymbol{HS}^{\dagger}) \\
\boldsymbol{Y} \otimes \boldsymbol{Y} & \boldsymbol{U} = CX_{10}(\boldsymbol{HS}^{\dagger} \otimes \boldsymbol{HS}^{\dagger})
\end{array}
$$

# Additional remarks

Here, the CNOT (CX10) operation appears for the following reason. Each Pauli measurement that does not include the identity matrix is equivalent up to a unitary to $\boldsymbol{Z} \otimes \boldsymbol{Z}$. The eigenvalues $\boldsymbol{Z} \otimes \boldsymbol{Z}$ of only depend on the parity of the qubits that comprise each computational basis vector, and the controlled-not operations serve to compute this parity and store it in the first bit. Then once the first bit is measured, you can recover the identity of the resultant half-space, which is equivalent to measuring the Pauli operator.

# Reducing space

Also, while it can be tempting to assume that measuring $\boldsymbol{Z} \otimes \boldsymbol{Z}$ is the same as sequentially measuring $\boldsymbol{Z} \otimes \boldsymbol{I}$ and then $\boldsymbol{I} \otimes \boldsymbol{Z}$, this assumption would be false. The reason is that measuring $\boldsymbol{Z} \otimes \boldsymbol{Z}$ projects the quantum state into either the $+1$ or $-1$ eigenstate of these operators. Measuring $\boldsymbol{Z} \otimes \boldsymbol{I}$ and then $\boldsymbol{I} \otimes \boldsymbol{Z}$ projects the quantum state vector first onto a half space of $\boldsymbol{Z} \otimes \boldsymbol{I}$ and then onto a half space of $\boldsymbol{I} \otimes \boldsymbol{Z}$. As there are four computational basis vectors, performing both measurements reduces the state to a quarter-space and hence reduces it to a single computational basis vector.

# Lipkin model

We will study a schematic model (the Lipkin model, see Nuclear Physics **62** (1965) 188), for the interaction among 2 and more fermions that can occupy two different energy levels.
In project 1 we consider a two-fermion case and a four-fermion case. For four fermions, the case we consider in the examples here, each levels has degeneration $d = 4$, leading to different total spin values. The two levels have quantum numbers $\sigma = \pm 1$, with the upper level having $2\sigma = +1$ and energy $\varepsilon_1 = \varepsilon/2$. The lower level has $2\sigma = -1$ and energy $\varepsilon_2 = -\varepsilon/2$. That is, the lowest single-particle level has negative spin projection (or spin down), while the upper level has spin up. In addition, the substates of each level are characterized by the quantum numbers $p = 1, 2, 3, 4$.

# Four fermion case

We define the single-particle states (for the four fermion case which we will work on here)

$$|u_{\sigma=-1,p}\rangle = a^{\dagger}_{-p}|0\rangle \qquad |u_{\sigma=1,p}\rangle = a^{\dagger}_{+p}|0\rangle.$$

The single-particle states span an orthonormal basis.

# Hamiltonian

The Hamiltonian of the system is given by

$$
\begin{aligned}
\hat{H} = &\quad \hat{H}_0 + \hat{H}_1 + \hat{H}_2 \\
\hat{H}_0 = &\quad \tfrac{1}{2}\varepsilon \textstyle\sum_{\sigma,p} \sigma a^{\dagger}_{\sigma,p} a_{\sigma,p} \\
\hat{H}_1 = &\quad \tfrac{1}{2} V \textstyle\sum_{\sigma,p,p'} a^{\dagger}_{\sigma,p} a^{\dagger}_{\sigma,p'} a_{-\sigma,p'} a_{-\sigma,p} \\
\hat{H}_2 = &\quad \tfrac{1}{2} W \textstyle\sum_{\sigma,p,p'} a^{\dagger}_{\sigma,p} a^{\dagger}_{-\sigma,p'} a_{\sigma,p'} a_{-\sigma,p}
\end{aligned}
$$

where $V$ and $W$ are constants. The operator $H_1$ can move pairs of fermions while $H_2$ is a spin-exchange term. The latter moves a pair of fermions from a state $(p\sigma, p' - \sigma)$ to a state $(p - \sigma, p'\sigma)$.

# Quasispin operators

We are going to rewrite the above Hamiltonian in terms of so-called quasispin operators

$$\begin{aligned}
\hat{J}_+ &= \sum_p a^\dagger_{p+} a_{p-} \\
\hat{J}_- &= \sum_p a^\dagger_{p-} a_{p+} \\
\hat{J}_z &= \frac{1}{2} \sum_{p\sigma} \sigma a^\dagger_{p\sigma} a_{p\sigma} \\
\hat{J}^2 &= J_+ J_- + J_z^2 - J_z
\end{aligned}$$

These operators obey the commutation relations for angular momentum.

# Including the number operator

We can in turn express $\hat{H}$ in terms of the above quasispin operators and the number operator

$$\hat{N} = \sum_{p\sigma} a^{\dagger}_{p\sigma} a_{p\sigma}.$$

# Final Hamiltonian

Using the quasispin operators we can rewrite the Hamiltonian as

$$H = \varepsilon J_z + \frac{1}{2} V \left( J_+^2 + J_-^2 \right) + \frac{1}{2} W \left( -N + J_+ J_- + J_- J_+ \right). \qquad (1)$$

# Original Lipkin Hamiltonian

The Lipkin model Hamiltonian in terms of quasispin operators reads

$$H = \varepsilon J_z + \frac{1}{2} V \left( J_+^2 + J_-^2 \right) + \frac{1}{2} W \left( -N + J_+ J_- + J_- J_+ \right).$$

We use the standard commutation relations

$$[J_+, J_-] = 2J_z,$$

and the Casimir operator

$$J^2 = J_z^2 + \frac{1}{2}(J_+ J_- + J_- J_+).$$

# More compact expression

From this we obtain

$$J_+J_- = J^2 - J_z(J_z - 1), \qquad J_-J_+ = J^2 - J_z(J_z + 1).$$

Adding the two expressions:

$$J_+J_- + J_-J_+ = 2(J^2 - J_z^2).$$

# Simplified Hamiltonian

Using

$$J_+J_- + J_-J_+ = 2(J^2 - J_z^2)$$

and insert into the Hamiltonian:

$$H = \varepsilon J_z + \frac{1}{2}V(J_+^2 + J_-^2) + \frac{1}{2}W\left(-N + 2(J^2 - J_z^2)\right).$$

This gives

$$\boxed{H = \varepsilon J_z + \frac{V}{2}(J_+^2 + J_-^2) + W(J^2 - J_z^2) - \frac{WN}{2}.}$$

# Physical interpretation

- The Hamiltonian is now expressed in terms of:
  - Linear term: $\varepsilon J_z$
  - Pairing term: $J_+^2 + J_-^2$
  - Quadratic term: $J_z^2$

  *Casimir term: $J^2$
- The constant $-\frac{WN}{2}$ only shifts the spectrum.

# Fixed quasispin sector ($j = N/2$)

In the Lipkin model we typically work in a fixed irreducible representation:

$$J^2 = j(j+1), \qquad j = \frac{N}{2}.$$

Thus

$$H = \varepsilon J_z + \frac{V}{2}(J_+^2 + J_-^2) - WJ_z^2 + Wj(j+1) - \frac{WN}{2}.$$

# Final simplified form

Using $N = 2j$, the constant simplifies:

$$Wj(j+1) - \frac{WN}{2} = Wj^2.$$

## Final result

$$\boxed{H = \varepsilon J_z + \frac{V}{2}(J_+^2 + J_-^2) - WJ_z^2 + Wj^2.}$$

The term $Wj^2$ is a constant shift and can often be omitted.

# Final Hamiltonian matrix

These five states can in turn be used as computational basis states in order to define the Hamiltonian matrix to be diagonalized. The matrix elements are given by $\langle J, J_z | H | J', J_z' \rangle$. The Hamiltonian is hermitian and we obtain after all this labor of ours

$$H_{J=2} = \begin{bmatrix} -2\varepsilon & 0 & \sqrt{6}V & 0 & 0 \\ 0 & -\varepsilon + 3W & 0 & 3V & 0 \\ \sqrt{6}V & 0 & 4W & 0 & \sqrt{6}V \\ 0 & 3V & 0 & \varepsilon + 3W & 0 \\ 0 & 0 & \sqrt{6}V & 0 & 2\varepsilon \end{bmatrix} \qquad (2)$$

## Quantum computing and solving the eigenvalue problem for the Lipkin model

We turn now to a simpler variant of the Lipkin model without the $W$-term and a total spin of $J = 1$ only as maximum value of the spin. This corresponds to a system with $N = 2$ particles (fermions in our case). Our Hamiltonian is given by the quasispin operators (see below)

$$\hat{H} = \epsilon \hat{J}_z - \frac{1}{2} V (\hat{J}_+ \hat{J}_+ + \hat{J}_- \hat{J}_-).$$

As discussed earlier the quasispin operators act like lowering and raising angular momentum operators.

With these properties we can calculate the Hamiltonian matrix for the Lipkin model by computing the various matrix elements

$$\langle J J_z | H | J J_z' \rangle, \tag{3}$$

where the non-zero elements are given by

$$\begin{aligned} \langle J J_z | H | J J_z' \rangle &= \epsilon J_z \\ \langle J J_z | H | J J_z' \pm 2 \rangle &= \langle J J_z \pm 2 | H | J J_z' \rangle \\ &= -\frac{1}{2} V C \end{aligned}$$

# Plans for next week

1. We will focus mainly on setting up the quantum circuit for the Lipkin model with $J = 1$ and $J = 2$, rewriting the Lipkin model in terms of Pauli matrices
2. Thereafter, if we get time we start with quantum Fourier transforms and discussions of Quantum Phase estimation algorithm